

GRAPH NEURAL NETWORKS FOR COARSE-GRAINED PROTEIN STRUCTURE PREDICTION FROM AMINO ACID SEQUENCE

Sergi Cases Alonso
Martí Pascual Barluenga

sergi.cases01@estudiant.upf.edu
marti.pascual01@estudiant.upf.edu

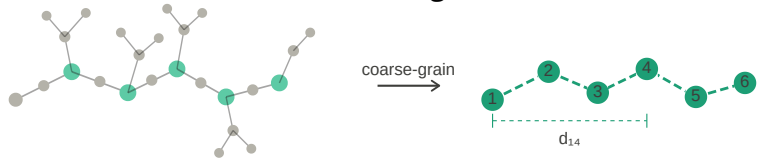


MOTIVATION

Can a simple neural network pipeline predict a coarse-grained 3D protein structure from amino acid sequence alone, using fewer resources than state-of-the-art methods like AlphaFold?

INTRODUCTION

Proteins fold into 3D shapes that determine their function. Experimental determination (X-ray, cryo-EM) is accurate but costly. We predict structure from sequence alone using a coarse-grained C α representation (one atom per residue), reducing the target to an N $\times$ N distance matrix predicted by a Transformer-Graph Neural Network (GNN) pipeline and reconstructed into 3D coordinates via Multidimensional Scaling (MDS).



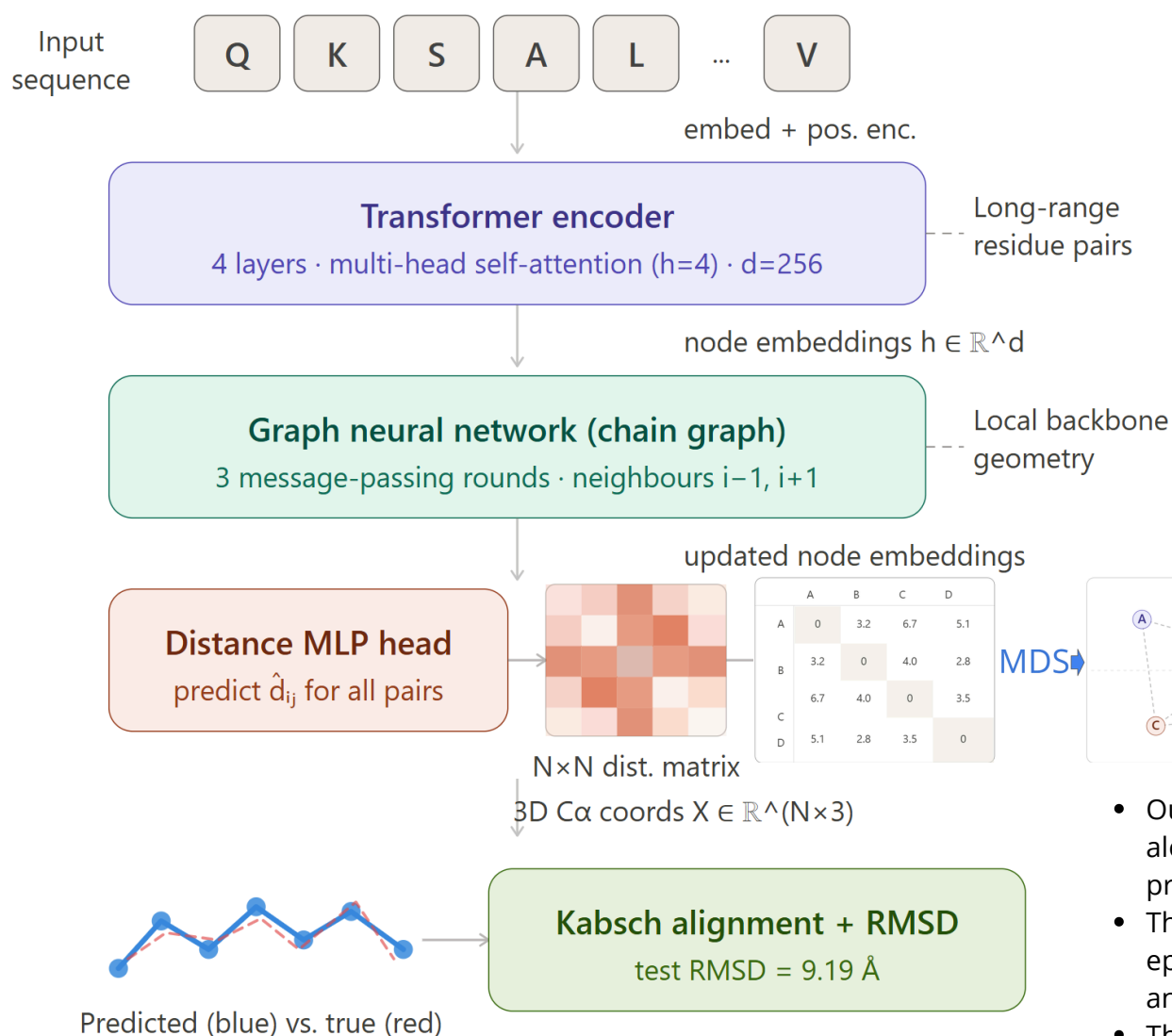
METHODS

Data and Preprocessing

Training data has 15,718 protein chains from the Protein Data Bank (PDB). The dataset is split 80/10/10 into train, validation, and test sets.

Each chain is parsed to extract the amino acid sequence and C α coordinates. The pairwise C α distance matrix is computed as the prediction target.

Pipeline



Loss function

$$\mathcal{L} = \mathcal{L}_{\text{MSE}} + \lambda \mathcal{L}_{\text{bond}}$$

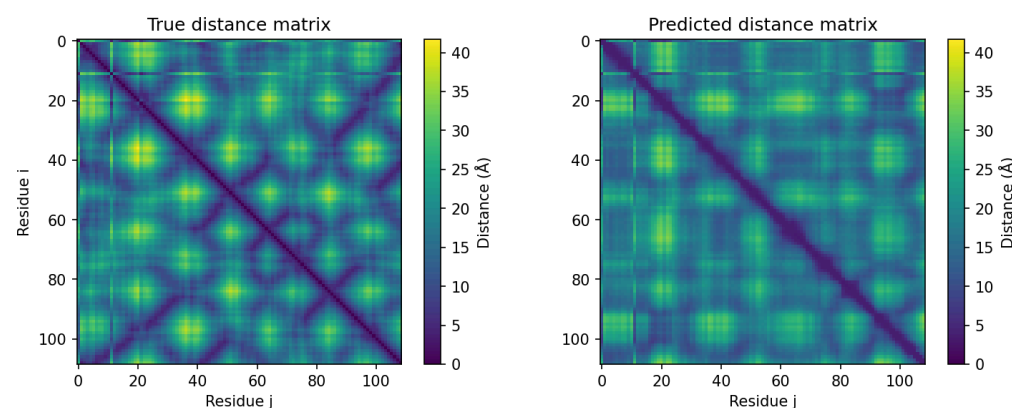
$\mathcal{L}_{\text{mse}}$ penalises errors over all residue pairs and $\mathcal{L}_{\text{bond}}$ enforces the 3.8 Å peptide-bond geometry between consecutive residues, with $\lambda=1.0$.

RESULTS AND DISCUSSION

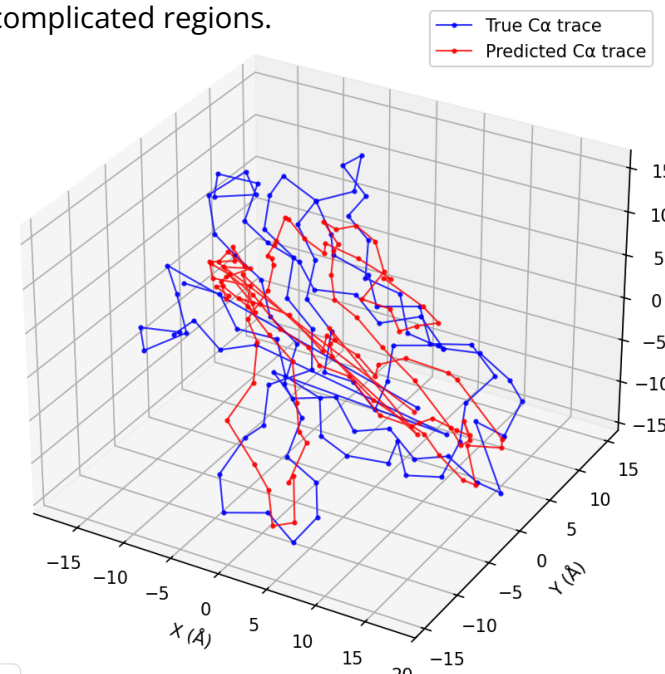
- The model was evaluated on 1,573 held-out PDB chains and obtained these results:

Mean test RMSD	Mean-distance baseline RMSD
9.19 Å	13.97 Å

- The model recovers the broad block structure of the map, including the near-diagonal short-distance band and several off-diagonal patterns.



- Reconstructed 3D C α backbone against the experimental structure (RMSD = 6.25 Å). The overall fold topology is captured, with larger deviations in complicated regions.



CONCLUSIONS

- Our pipeline predicts coarse-grained C α structure from sequence alone, achieving a mean test RMSD of 9.19 Å on 1,573 unseen proteins, well below the 13.97 Å mean-distance baseline.
- The loss curve shows convergence without overfitting across 100 epochs, validating the combination of MSE loss, bond constraint, and L2 regularisation.
- The 3D backbone reconstruction for 10ZO_A (RMSD = 6.25 Å) shows that the overall fold is captured.
- Limitations:** MDS amplifies distance prediction errors into coordinate space, inflating RMSD beyond what the distance MAE alone suggests.
- Future work.** Replacing learned embeddings with pre-trained protein language model representations could improve the results significantly.

REFERENCES

- J. Jumper, R. Evans, A. Pritzel, et al., "Highly accurate protein structure prediction with AlphaFold," Nature, vol. 596, pp. 583–589, 2021.
- A. Vaswani, N. Shazeer, N. Parmar, et al., "Attention is all you need," Advances in Neural Information Processing Systems, vol. 30, pp. 5998–6008, 2017.
- T. N. Kipf and M. Welling, "Semi-supervised classification with graph convolutional networks," International Conference on Learning Representations, 2017.
- T. Wu, Z. Guo, J. Hou, and J. Cheng, "DeepDist: Real-value inter-residue distance prediction with deep residual convolutional network," BMC Bioinformatics, vol. 22, pp. 1–17, 2021.